

THREE WAVE INTERACTION AND DYNAMIC WAVE TURBULENCE

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1. INTRODUCTION

Our motivation stems from the Dzhanibekov effect, better known as the tennis racket theorem. This is a popular phenomenon within classical mechanics that describes the behavior of a rotating rigid body when the moments of inertia corresponding to the principle axes are all distinct. As will be seen later on in the paper, this induces instability along one axis, and stability along the other two. This can even be demonstrated at home with a tennis racket. Flipping it so that the handle rotates will result in a stable rotation. Likewise, flipping the racket so that the net is perpendicular to the ground will also result in a stable rotation. However, if you were to flip the tennis racket such that the net is parallel to the ground, you would find that the rotational path of the tennis racket is wobbly, and this is exactly the phenomenon that kick-started this project.

2. BACKGROUND

With our research, we aim to analyze the dynamics of dispersion within systems of three and four wave partial differential equations, such as the topology of cascading clusters formed by resonance triads and their inherent interactions. We begin by analyzing the system of ordinary differential equations known as Euler's Rigid Body Equations.

Definition 2.1. Euler's Rigid Body Equations: Euler's Rigid Body Equations are a system of three ordinary differential equations which depict the rotation of a rigid body about its three axes. This system is of the form:

$$\begin{aligned} \dot{\omega}_1 &= \frac{(I_3 - I_2)}{I_1} \omega_2 \omega_3 \\ \dot{\omega}_2 &= \frac{(I_1 - I_3)}{I_2} \omega_3 \omega_1 \\ \dot{\omega}_3 &= \frac{(I_2 - I_1)}{I_3} \omega_1 \omega_2 \end{aligned} \tag{1.1}$$

where $I_1 > I_2 > I_3$ represent the moments of inertia about each axis.

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Definition 2.2. Differential Equation: [1] A differential equation is a class of function that describes the rate at which something occurs.

A keen eye might notice that the sign of $\dot{\omega}_2$ in (1.1) is positive given the relative equality of the , whereas the other two are negative. This tells us that the rotation about the ω_2 axis will initially dominate. Although this behavior will cascade through the other two equations, as they involve a ω_2 term. This rotational instability will modulate between ω_2 and the pair ω_1 and ω_3 . Such a phenomenon can be defined as modular instability.

Definition 2.3. Modulation Instability: [5] Modulation instability is a phenomenon, catalyzed by an initial excitation within a system. Following this excitation is instability that modulates through a system that was previously at rest.

It is modular instability that we aim to analyze within wave systems. Modular instability can be displayed mathematically as a dynamical system for the time evolution of the amplitudes of certain waves. How do we know which waves exhibit this behavior? Well, we can perform what is known as a Fourier transform to obtain a function relating frequency and wavenumber, known as a dispersion relation.

Definition 2.4. Dispersion Relation: [2] A dispersion relation is a solvability condition on the governing wave equation that links frequency and wavenumber. This is defined as a function of the form:

$$\omega = \omega(k) : \frac{d^2\omega}{dk^2} \neq 0$$

It is with this dispersion relation that we can essentially filter out any system that won't exhibit modular instability and form resonant triads. Such a filter is referred to as the resonance condition.

Definition 2.5. Resonance Condition [3]: A resonance condition fulfills the following for a dispersion function $\omega_i(k_i)$, with ω representing frequency and k referencing wavenumber:

$$(1.2) \quad \begin{aligned} k_1 + k_2 &= k_3 \\ \omega_1 + \omega_2 &= \omega_3 \end{aligned}$$

Definition 2.6. Resonance Triad: [4] A resonance triad is a set of three points, (ω_i, k_i) , known as wave modes, that fulfill the the resonance condition.

Every triad can be plotted as a cluster, where each element represents what will come to be known as a wave mode. These wave modes are ordered pairs (k_i, ω_i) of two types: A-modes and P-modes. The A-mode will always have the largest frequency out of the modes in a triad, like (k_3, ω_3) in (1.2) . These modes cause the initial excitement within the system. The interconnectedness of these modes dictate the topology of the clusters. Each wavemode represents amplitude, and we can also analyze the time evolution of energy in a system by a primary dynamical system.

Definition 2.7. Primary Dynamical System: [include before talking about multiple triads] A primary dynamical system is a system of ordinary differential equations that describe the time evolution of amplitude within a resonance triad:

$$(1.3) \quad \begin{aligned} \dot{B}_1 &= -ZB_2B_3 \\ \dot{B}_2 &= ZB_1B_3 \\ \dot{B}_3 &= ZB_1B_2 \end{aligned}$$

where each B_i is a wavemode.

A single wave mode can belong to multiple resonance triads, which leads towards the interconnectedness of triads. The interconnectedness of these triads by shared modes dictate the topology of what clusters. The structure of these clusters can take multiple forms. Such an example 2.1 is a PP-butterfly cluster, which is two triads connected by a shared mode that acts as a P-mode within both triads. Expanding (1.3) to a system seen in 2.1 where the shared P mode is $\dot{B}_3$ yields the following system:

$$(1.4) \quad \begin{aligned} \dot{B}_1 &= -ZB_2B_3 \\ \dot{B}_2 &= ZB_1B_3 \\ \dot{B}_3 &= ZB_1B_2 + ZB_4B_5 \\ \dot{B}_4 &= ZB_3B_5 \\ \dot{B}_5 &= -ZB_3B_4 \end{aligned}$$

As can be seen, $\dot{B}_3$ receives energy from the unstable $\dot{B}_1$ and $\dot{B}_5$ modes, which, when plotted like in 2.2, show the amplification in magnitude that we would expect from looking at the combination of terms in (1.4). From this, we wanted to look further into the dynamics of these clusters and better understand their structure and behavior both numerically and analytically.

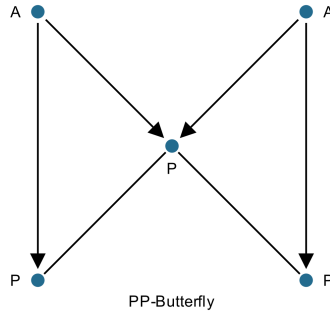


FIGURE 2.1

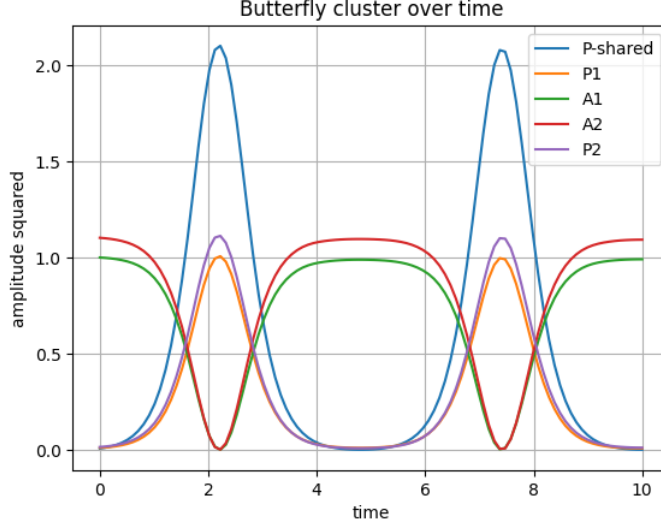


FIGURE 2.2

3. RESULTS

3.1. Numerical Results. Our goal here was to numerically model clusters in Python. To begin, we wrote some code that would initialize a chosen wavenumber domain. For each element (k_1) in the set, our code would solve the dispersion relation $\omega(k_1)$, and subsequently generate some $(k_2, \omega(k_2)), (k_3, \omega(k_3))$ such that the resonance condition is fulfilled. These tuples are then saved to a set. With this data, we aimed to model the clusters.

3.1.1. Plotting Approach. The path we took to represent these clusters was by plotting the primary dynamical systems. We simply constructed a system based off the cluster, and used the `solve_ivp` function from the SciPy API to generate arrays of solution data that we used to plot the graphs. With this, we were able to plot chains of clusters of similar structure to 3.1. We were then interested in what different types of forcing/damping would inhibit in terms of behavior. To do this, we first constructed a 7-chain system. Next, we applied a damping term to the last P-mode in the chain, and gave it a relatively high initial condition. The primary dynamical system of such a system with two triads is as follows:

$$\begin{aligned}
 \dot{B}_1 &= -ZB_2B_3 \\
 \dot{B}_2 &= ZB_1B_3 \\
 \dot{B}_3 &= ZB_1B_2 - ZB_4B_5 \\
 \dot{B}_4 &= ZB_3B_5 \\
 \dot{B}_5 &= ZB_3B_4 - D
 \end{aligned}
 \tag{1.5}$$

where D is a linear damping term.

The result 3.2 is a system where the dangling P-modes reach a non-zero equilibrium. This result was surprising, as we would expect the equilibrium solutions for these P-modes to be 0. To investigate this further, we turned to analytical means.

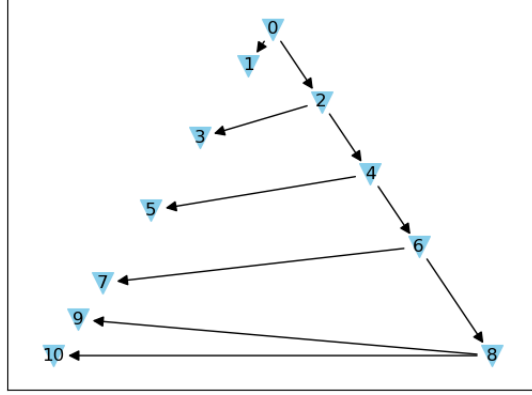


FIGURE 3.1. Simple directed graph of a 5-chain cluster

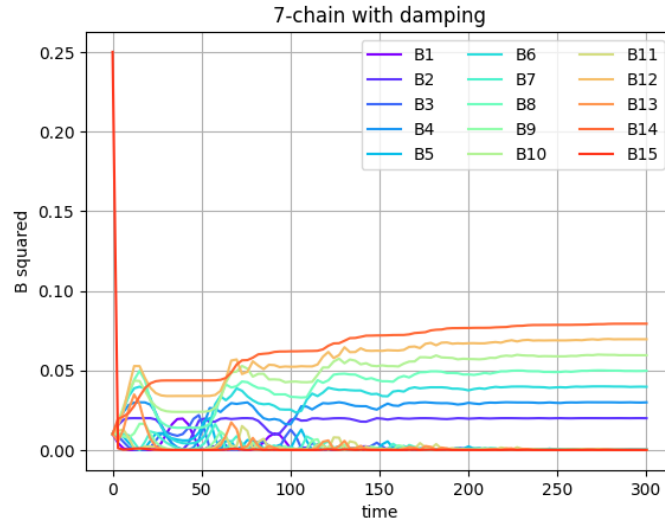


FIGURE 3.2

3.2. Analytical Results. To analyze the behavior we saw in 3.2, we first examined the equilibrium points of a 1-chain system, whose system would

be:

$$\begin{aligned}
 \dot{B}_1 &= ZB_2B_3 \\
 \dot{B}_2 &= ZB_1B_3 \\
 \dot{B}_3 &= ZB_1B_2 - DB_3
 \end{aligned}
 \tag{1.6}$$

We can then see that $B_3 = 0$ must hold, so that means either $B_1 = 0$ and B_2 can basically be anything, or $B_2 = 0$ and B_1 can be basically anything. We now turn to the Jacobian Matrix for (1.6):

$$J_3 = \begin{bmatrix} 0 & -ZB_3 & -ZB_2 \\ ZB_3 & 0 & ZB_1 \\ ZB_2 & ZB_1 & 0 \end{bmatrix}
 \tag{1.7}$$

and solve for the eigenvalues of the system and then analyze the sign of their real parts. We found $\begin{bmatrix} 0 \\ B_2 \\ 0 \end{bmatrix}$ to be stable with a negative real part and $\begin{bmatrix} B_1 \\ 0 \\ 0 \end{bmatrix}$ to be unstable with a positive real part. Extending this process to a 2-chain like (1.5), we first plotted 3.3. One can see that $[0 \ B_2 \ 0 \ B_4 \ 0]^T$ this will be a stable equilibrium point. Then, by solving for the Jacobian, we find that the other equilibrium points are $[B_1 \ 0 \ 0 \ B_4 \ 0]^T$ and $[0 \ 0 \ B_3 \ 0 \ 0]^T$. We were able to determine, using the Descartes Sign Theorem, that both of these equilibrium points will have a positive real part.

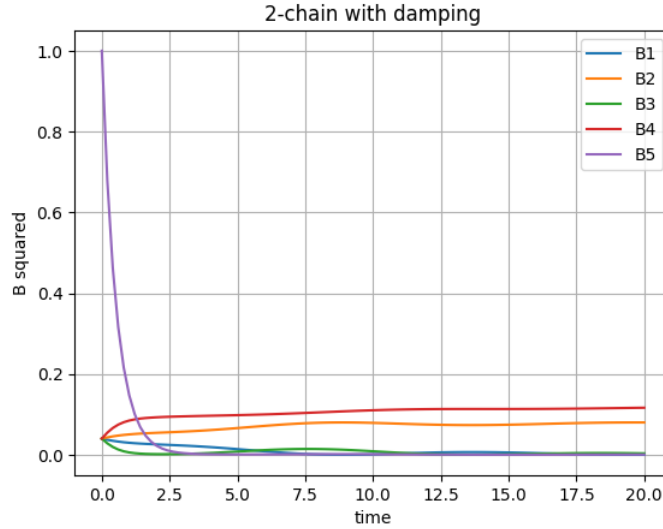


FIGURE 3.3

4. LOOKING FORWARD

Thus far we have successfully produced numerical models of these dynamical systems, and have constructed graphical representation of clusters. From there we analyzed the equilibrium solutions of an n-chain with a damped P-mode containing the initial energy. We would like to further generalize this, and experiment more with damping/forcing to induce a reverse cascade. Furthermore we could also perform statistical analysis on cluster topological structure given differing dispersion functions.

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